MD. SADBIN HOSSAIN

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LinkedIn | GitHub | Portfolio

Research Profile

Undergraduate Computer Science and Engineering student at American International University-Bangladesh (AIUB), with current manuscript work spanning efficient and explainable deep learning for medical-image classification and secure electronic health-record (EHR) sharing. Interested in reproducible evaluation, model reliability under dataset shift, and privacy-aware healthcare systems. Seeking to contribute as a student researcher through careful experimentation, literature review, technical implementation, and collaborative research.

Education

American International University-Bangladesh (AIUB) Dhaka, Bangladesh
B.Sc. in Computer Science and Engineering; ***CGPA: 3.67/4.00*** *Expected 2027*
Current semester: 10th (undergraduate).

Research Interests

Efficient Deep Learning; Knowledge Distillation; Vision Transformers (ViT); Medical Image Analysis; Explainable AI (XAI); Model Robustness and Domain Shift; Federated Learning; Secure Healthcare Informatics, including post-quantum cryptography, blockchain, and EHR systems.

Research Manuscripts and Work in Progress

A Reproducible Lightweight Evaluation of Chest X-Ray Classification Under Cross-Dataset Shift
- Faculty supervisor: Dr. Kamruddin Md. Nur.
- Status: Submitted to ICCIT 2026; currently under conference review.

Lightweight Explainable Diabetic Foot Ulcer Classification via Knowledge Distillation
- Faculty supervisor: Dr. Muhammad Firoz Mridha.
- Status: Manuscript under faculty/supervisor review; conference submission is planned.

Post-Quantum Hybrid EHR Sharing Framework Using ML-KEM, ECC, Blockchain, and Off-Chain Storage
- Faculty supervisor: Md. Sajid Bin-Faisal.
- Status: Manuscript under faculty review; submission venue not yet specified.

Federated Learning for Healthcare/AI Applications (working title to be finalized)
- Faculty mentor: Dr. Abdus Salam (AIUB).
- Status: Manuscript in early development; target journal and scope are not yet finalized.

Selected Technical Projects

s4darc — Interactive Arch Linux Installer *2026–Present*
- Developed a shell-script-based interactive installer for Arch Linux with environment checks, guided disk and partition workflows, filesystem setup, system configuration, user creation, kernel selection, and bootloader setup for UEFI/BIOS systems.

s4dbox — Self-Hosted Seedbox Management Toolkit *2026–Present*
- Built a Bash terminal UI to install and manage media, cloud, torrent, automation, and remote-access applications, with system status/monitoring, tuning, and security commands for SSH, firewall, and Fail2ban; optional Nginx reverse-proxy support.

Travel Management System *Academic project*
- Developed a desktop application using C# Windows Forms with a MySQL database.

Life of an Office Worker in a Modern Urban Area *Academic project*
- Collaborated on a 2D computer-graphics animation implemented in C++ as a team project.

Technical Skills

Programming: C, C++, C#, Java, Python, Bash/Shell scripting

Systems: Linux, Arch Linux installation and configuration, command-line tooling, Windows

Networking and security project experience: SSH, firewall and Fail2ban controls, remote-access/VPN tool management, optional Nginx reverse proxy

Databases: MySQL (academic project)

Research and writing tools: LaTeX, Overleaf, Git, GitHub

Professional Development

Code to Cloud: Practical DevOps with AWS *2026*
- Workshop participation; hosted by AIUB Computer Club in collaboration with Cloudly Infotech Limited.